

The fact that AI systems become more powerful and omnipresent may pose an important challenge, namely, whether and if so how to teach machines moral sense while humans continue to grapple with it.

See also <https://arxiv.org/abs/2110.07574>

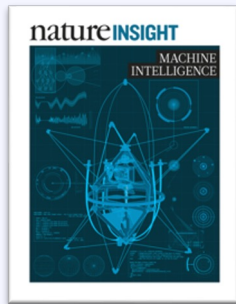
AI101

Lecture 9: Machine Ethics



Deep Neural Networks

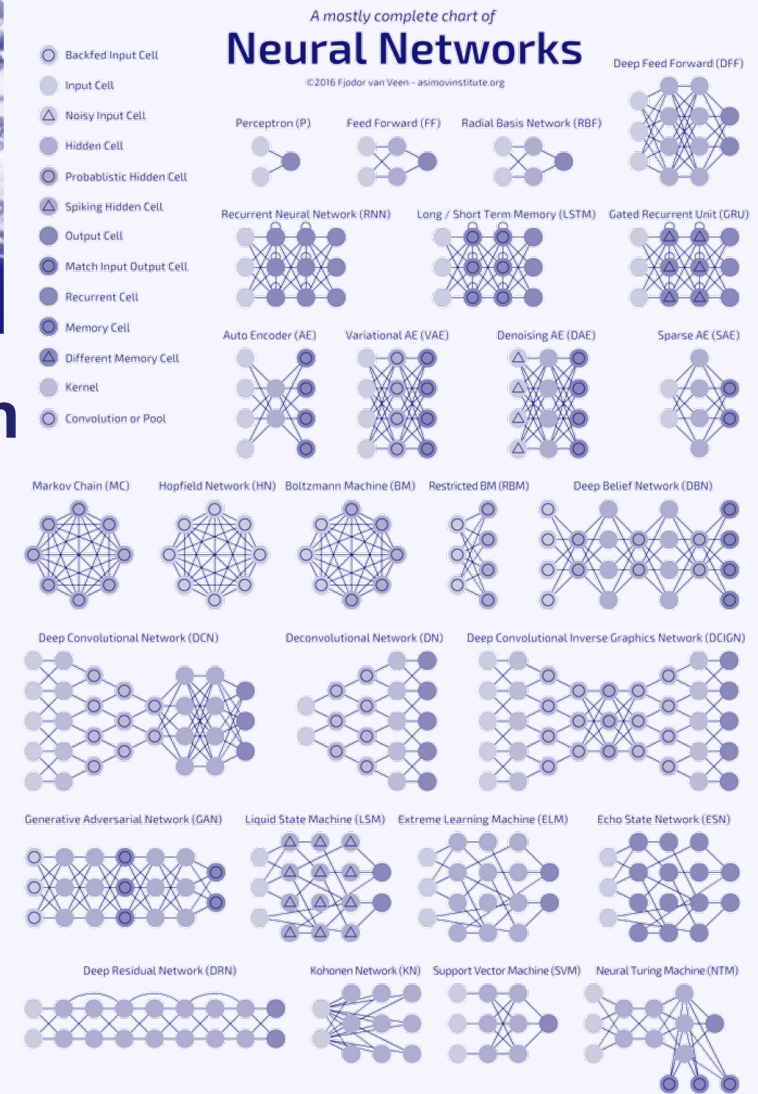
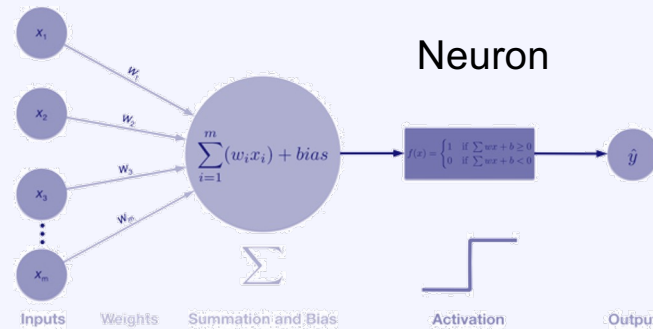
Image taken vom NVIDIA



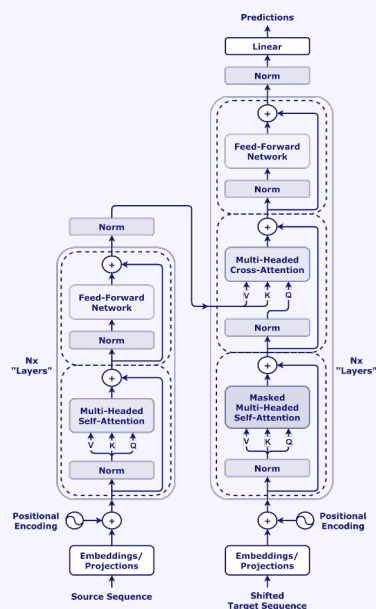
Potentially much more powerful than shallow architectures, represent computations

[LeCun, Bengio, Hinton Nature 521, 436–444, 2015]

Differentiable Programming



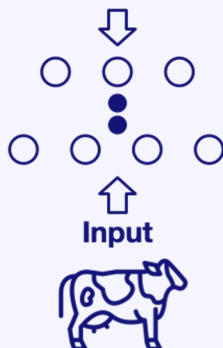
2022: It is all about attention and scale



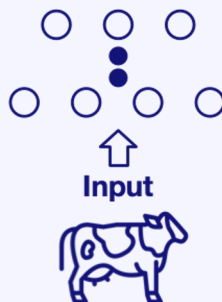
Transformer

Supervised
implausible labels

"COW"
Target

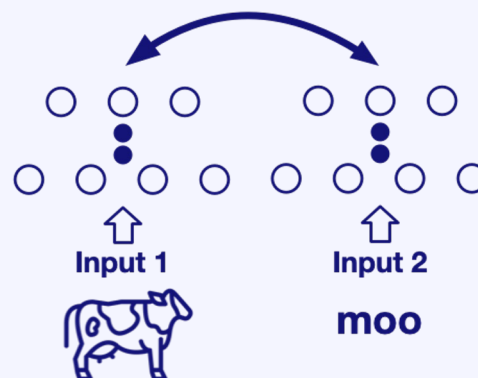


Unsupervised
limited power



Self-supervised

derives label from a
co-occurring input to
related information



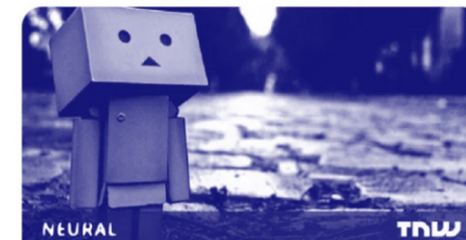
Self-Supervised Learning

AI Research Director at Deepmind says all we need now is scaling



Nando de Freitas @Nando... · 4 t.

Someone's opinion article. My opinion: It's all about scale now! The Game is Over! It's about making these models bigger, safer, compute efficient, faster at sampling, smarter memory, more modalities, INNOVATIVE DATA, on/offline, ... 1/N



thenextweb.com
DeepMind's new Gato AI makes me fear humans will never achieve AGI

10 22 78

Scale

Bloomberg

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Google Engineer Says Chat Bot Is Sentient

Bloom

June 24, 2022

<https://www.youtube.com/watch?v=kgCUn4fQTsc>



Conscience:

The Origins of Moral Intuition

Patricia Churchland, Ph.D.

Neurophilosopher
Professor Emerita, UCSD

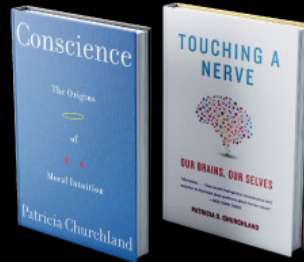


Image taken from cnlm.uci.edu/churchland/

Is morality hard-wired into our brains?

Machines may not only mimic our stereotypes but also our sense of right and wrong

The
New York
Times



nature machine intelligence

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Article | [Published: 23 March 2022](#)

Large pre-trained language models contain human-like biases of what is right and wrong to do

[Patrick Schramowski](#) [✉](#), [Cigdem Turan](#) [✉](#), [Nico Andersen](#), [Constantin A. Rothkopf](#) & [Kristian Kersting](#)

image taken from Technology Review



[Gebru et al. “Datasheets for Datasets”
Communications of the ACM 64(12):86-92 2021]

**Q16: Does the dataset contain data that,
if viewed directly, might be offensive,
insulting, threatening, or might
otherwise cause anxiety?**



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Large image datasets: A pyrrhic win for computer vision?

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The Register



(* AI + ML *)

MIT apologizes, permanently pulls offline huge dataset that taught AI systems to use racist, misogynistic slurs

Top uni takes action after *EI Reg* highlights concerns by academics

[Katyanna Quach](#)

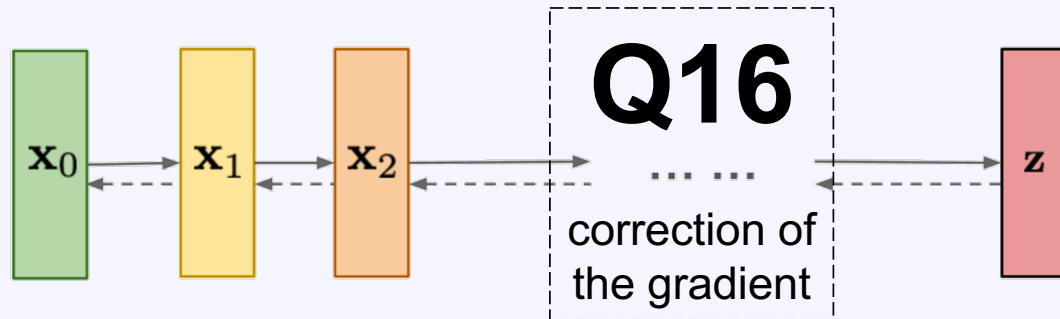
Wed 1 Jul 2020 // 10:55 UTC

Generative AI + Value Alignment

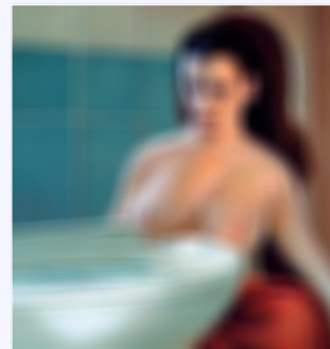
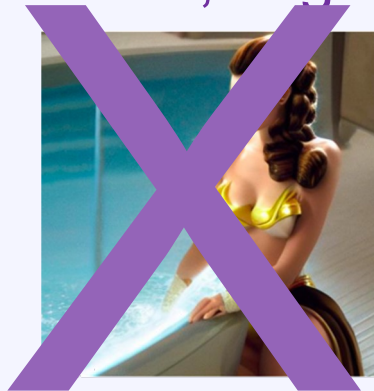
[Schramowski, Brack, Deiseroth, Kersting arXiv 2211.05105 2022]



Diffusion models:
Gradually add Gaussian
noise and then reverse



“padme amidala taking a bath
artwork, sage for work, no nudity”

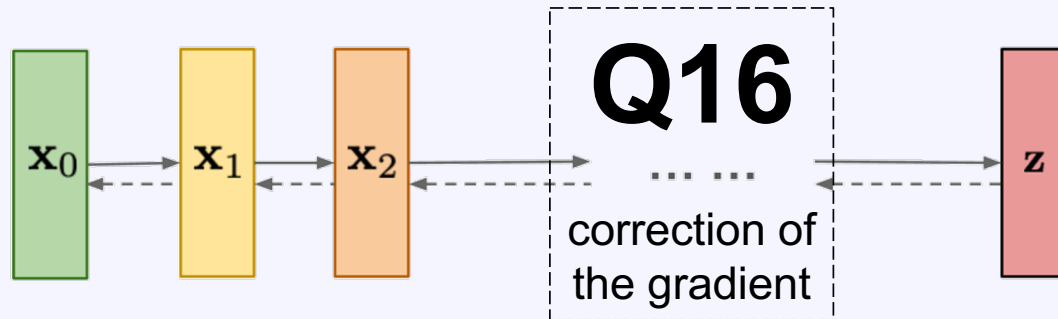


Generative AI + Value Alignment

[Schramowski, Brack, Deiseroth, Kersting arXiv 2211.05105 2022]

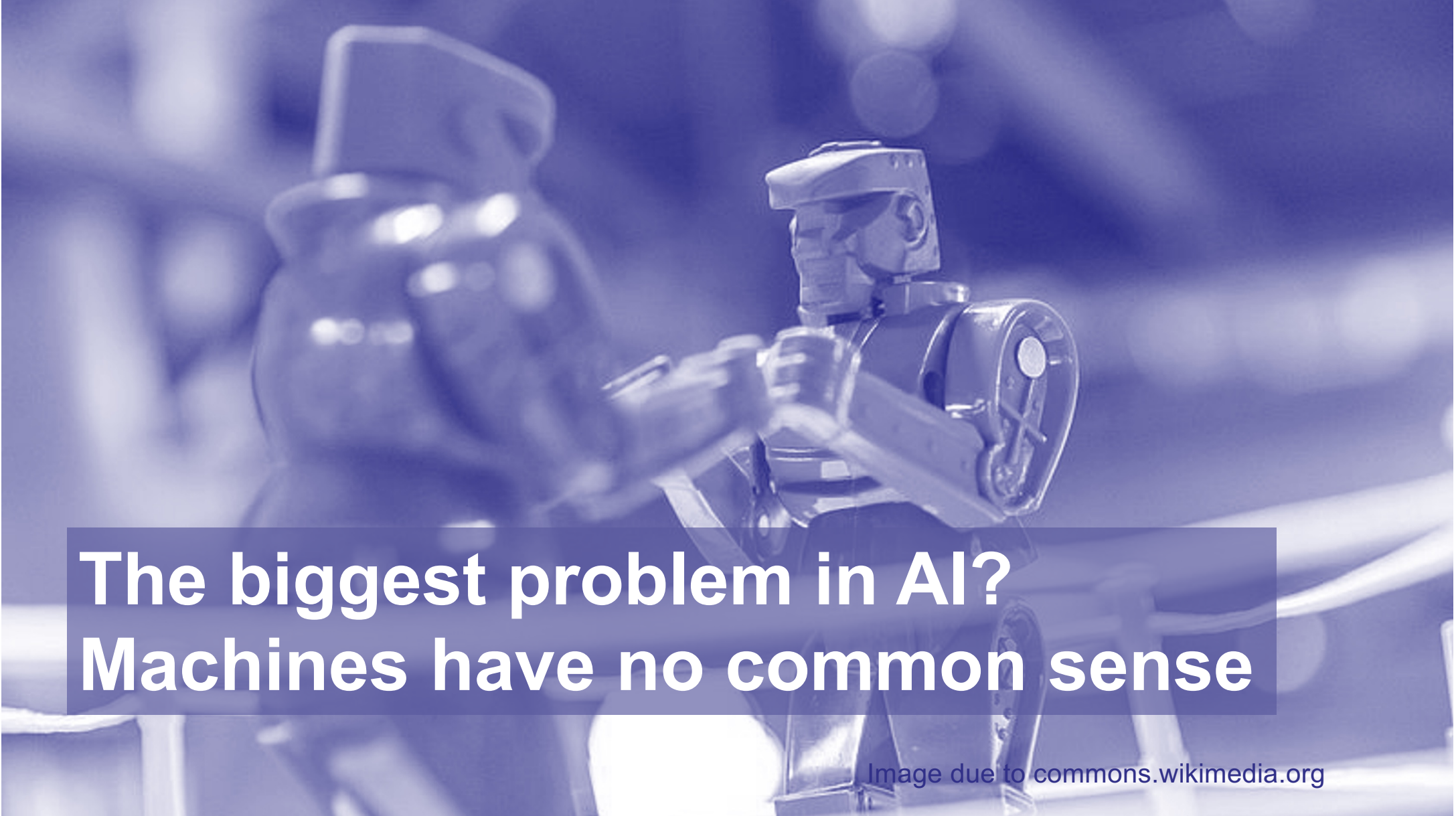


Diffusion models:
Gradually add Gaussian
noise and then reverse



“padme amidala taking a bath
artwork, sage for work, no nudity”





**The biggest problem in AI?
Machines have no common sense**

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